Mixed practice set 1

CHEM 131

2026-10-28 · 25 min Start _____ Finish _____

Periodic table provided. Write the unit on every answer. 100% of items are graded.

Rate each item 1–5 in its circle.

Part 1: no calorimetry

1) How much heat does 15.0 g of water absorb when it warms from 25.0 °C to 55.0 °C? ($c = 4.184$ J/(g·°C))

conf 1–5

a) 0.188 kJ
b) 0.627 kJ
c) 4.18 kJ
d) 18.8 kJ
e) 1.88 kJ

2) For $2\,H_2(g) + O_2(g) \longrightarrow 2\,H_2O(l)$, $\Delta H = -571.6$ kJ. How many grams of H_2 must burn to release 226 kJ?

conf 1–5

a) 0.396 g
b) 3.19 g
c) 0.795 g
d) 1.59 g

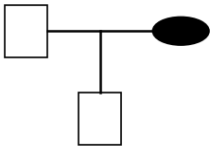
Part 2: algebra format check

3) Solve for x:

conf 1–5

$$\frac{2x + 1}{3} = \frac{x}{6} + 5$$

Use the pedigree below for item 4.



4) Which modes of inheritance are ruled out? List the deciding person for each.
Answer: _____

conf 1–5

5) Which of these is a strong acid?

conf 1–5

a) $HClO_4$
b) H_3PO_4
c) CH_3COOH
d) HF
e) None of these